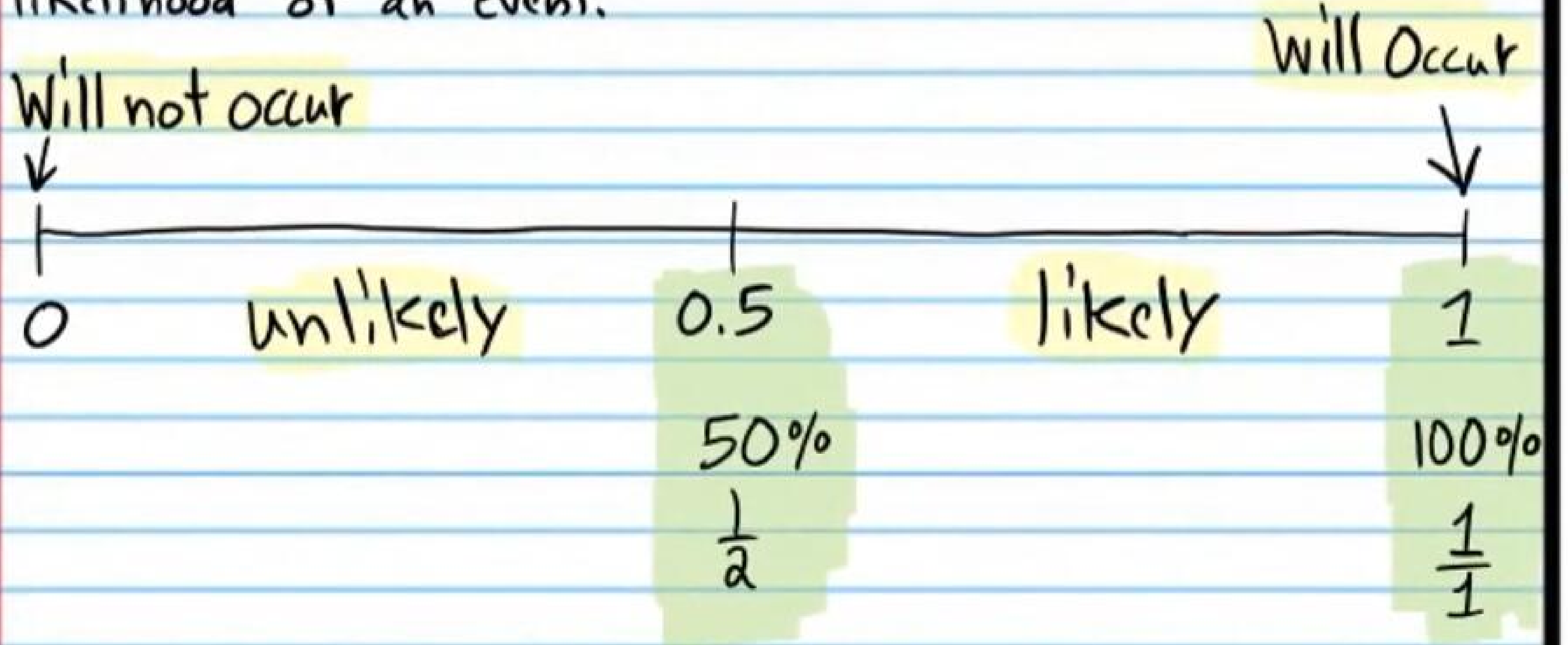
Name
Date
Course

Probability

Probability: A number from zero to one that indicates the likelihood of an event.



$$\text{Probability of an Event} = \frac{\text{\# of outcomes in the event}}{\text{Total \# of Outcomes possible}}$$

① In one week Ford Motor Company produces 207 trucks. 109 of them have cruise control, 28 of them have sunroofs, and 20 of them have both cruise control and sunroofs.

i) How many trucks have sunroofs but not cruise control?

$28 - 20 = \boxed{8}$

ii) How many trucks have cruise control but not sunroofs?

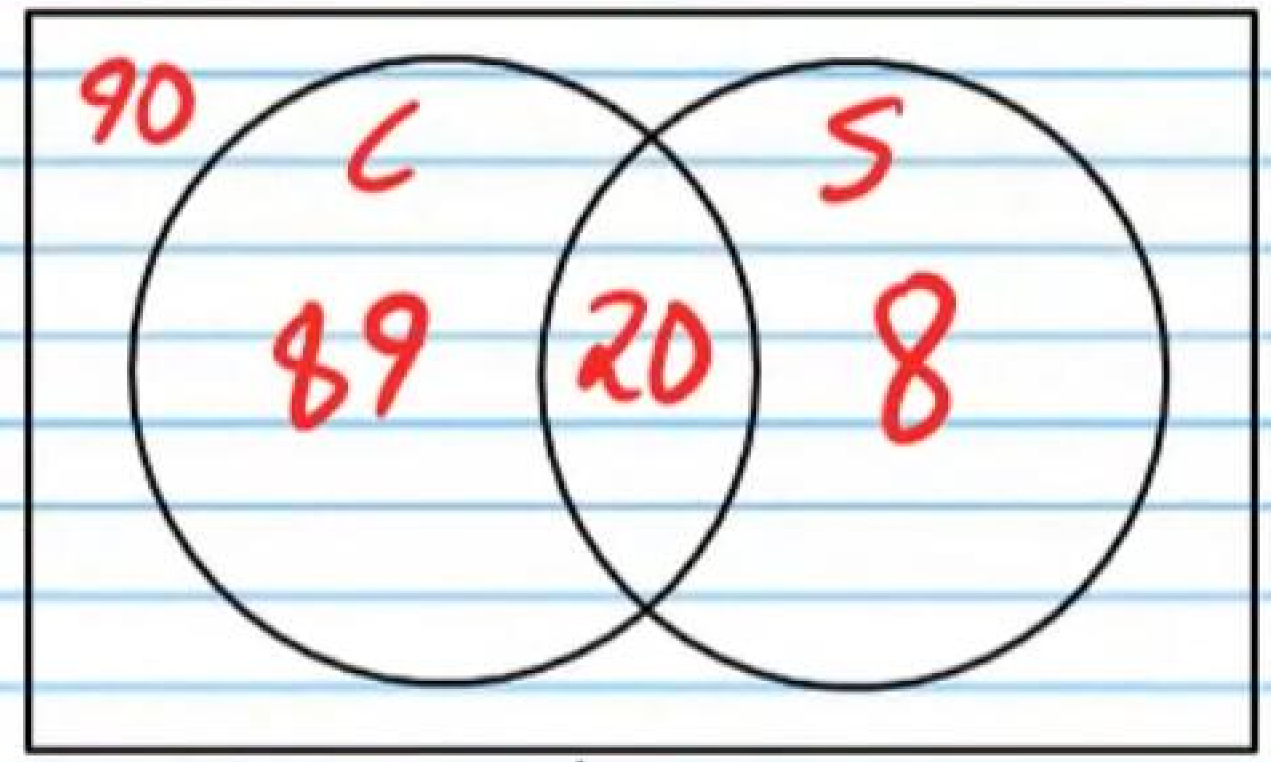
$109 - 20 = \boxed{89}$

iii) How many trucks have either sunroofs or cruise control?

$89 + 20 + 8 = \boxed{117}$

iv) How many trucks have neither sunroofs nor cruise control?

$207 - 117 = \boxed{90}$



v) What is the probability that a randomly selected truck will have cruise control?

$\boxed{\frac{109}{207}}$

vi) What is the probability that a randomly selected truck will have a sunroof?

$\boxed{\frac{28}{207}}$

vii) What is the probability that a randomly selected truck will have neither cruise control nor a sunroof?

$\frac{90}{207} = \frac{30}{69} = \boxed{\frac{10}{23}}$

viii) What is the probability that a randomly selected truck will have both cruise control and a sunroof?

$\boxed{\frac{20}{207}}$

ix) What is the probability that a randomly selected truck will have either cruise control or a sunroof?

$\frac{89+20+8}{207} = \frac{117}{207} = \frac{39}{69} = \boxed{\frac{13}{23}}$

Sample Space: Total # of outcomes possible.

$$\text{Probability of an Event} = \frac{\text{\# of outcomes in the event}}{\text{Total \# of Outcomes possible}}$$

$$= \frac{\text{\# of outcomes in the event}}{\text{Sample Space}}$$

② A family of 36 went on vacation in the mountains. 19 skied and 15 fished. 10 did both.

i) How many family members skied only?

$$19 - 10 = \boxed{9}$$

ii) How many family members fished only?

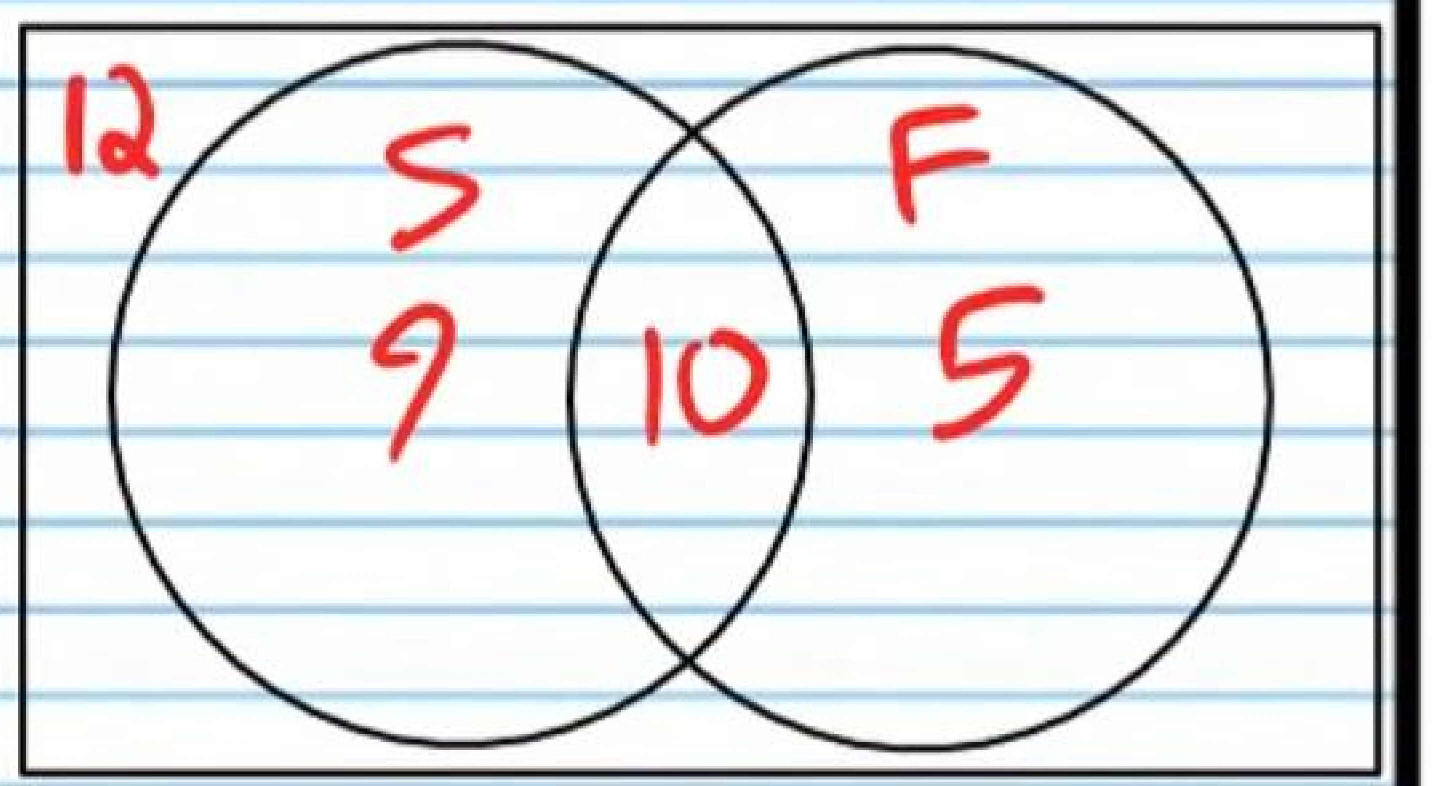
$$15 - 10 = \boxed{5}$$

iii) How many family members either fished or skied?

$$9 + 10 + 5 = \boxed{24}$$

iv) How many family members neither fished nor skied?

$$36 - 24 = \boxed{12}$$



v) What is the probability that a randomly selected family member skied?

$$\boxed{\frac{19}{36}}$$

vi) What is the probability that a randomly selected family member fished?

$$\frac{15}{36} = \boxed{\frac{5}{12}}$$

vii) What is the probability that a randomly selected family member neither skied nor fished?

$$\frac{12}{36} = \boxed{\frac{1}{3}}$$

viii) What is the probability that a randomly selected family member both skied and fished?

$$\frac{10}{36} = \boxed{\frac{5}{18}}$$

ix) What is the probability that a randomly selected family member either skied or fished?

$$\frac{9+10+5}{36} = \frac{24}{36} = \boxed{\frac{2}{3}}$$

③ There are 100 students in a school. 78 of them like peanut butter. 24 of them like jelly. 14 of them like both.

i) How many students like peanut butter but not jelly?

$78 - 14 = \boxed{64}$

ii) How many students like jelly but not peanut butter?

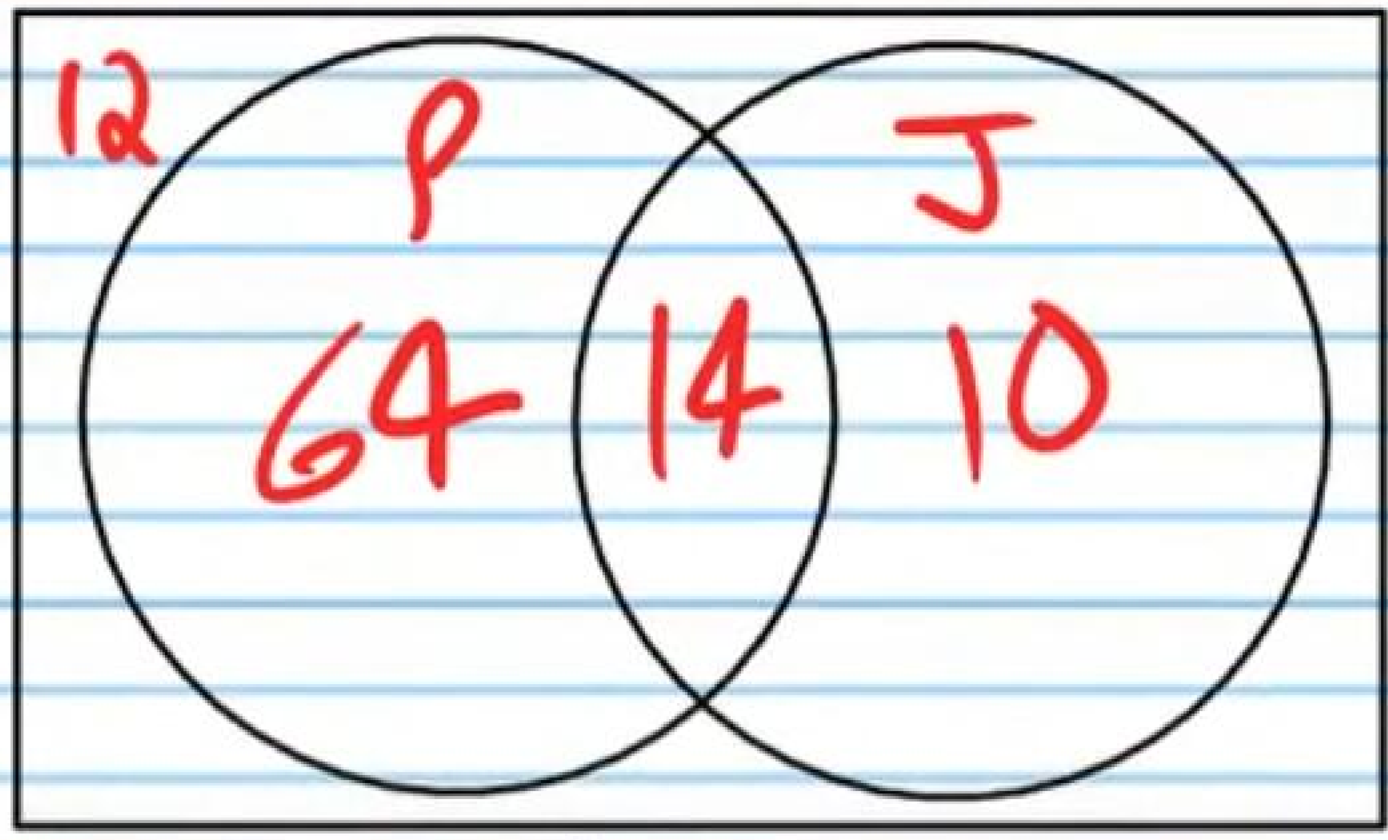
$24 - 14 = \boxed{10}$

iii) How many students like either peanut butter or jelly?

$64 + 14 + 10 = \boxed{88}$

iv) How many students do not like peanut butter or jelly?

$100 - 88 = \boxed{12}$



v) What is the probability that a randomly selected student will like peanut butter?

$$\frac{78}{100} = \boxed{\frac{39}{50}}$$

vi) What is the probability that a randomly selected student will like jelly?

$$\frac{24}{100} = \boxed{\frac{6}{25}}$$

vii) What is the probability that a randomly selected student will like neither peanut butter nor jelly?

$$\frac{12}{100} = \boxed{\frac{3}{25}}$$

viii) What is the probability that a randomly selected student will like both peanut butter and jelly?

$$\frac{14}{100} = \boxed{\frac{7}{50}}$$

ix) What is the probability that a randomly selected student will like either peanut butter or jelly?

$$\frac{64 + 14 + 10}{100} = \frac{88}{100} = \boxed{\frac{22}{25}}$$

④ There are 32 students in a class. 20 of them like math, 14 of them like art, and 8 of them like both.

i) How many students like math only?

$20 - 8 = \boxed{12}$

ii) How many students like art only?

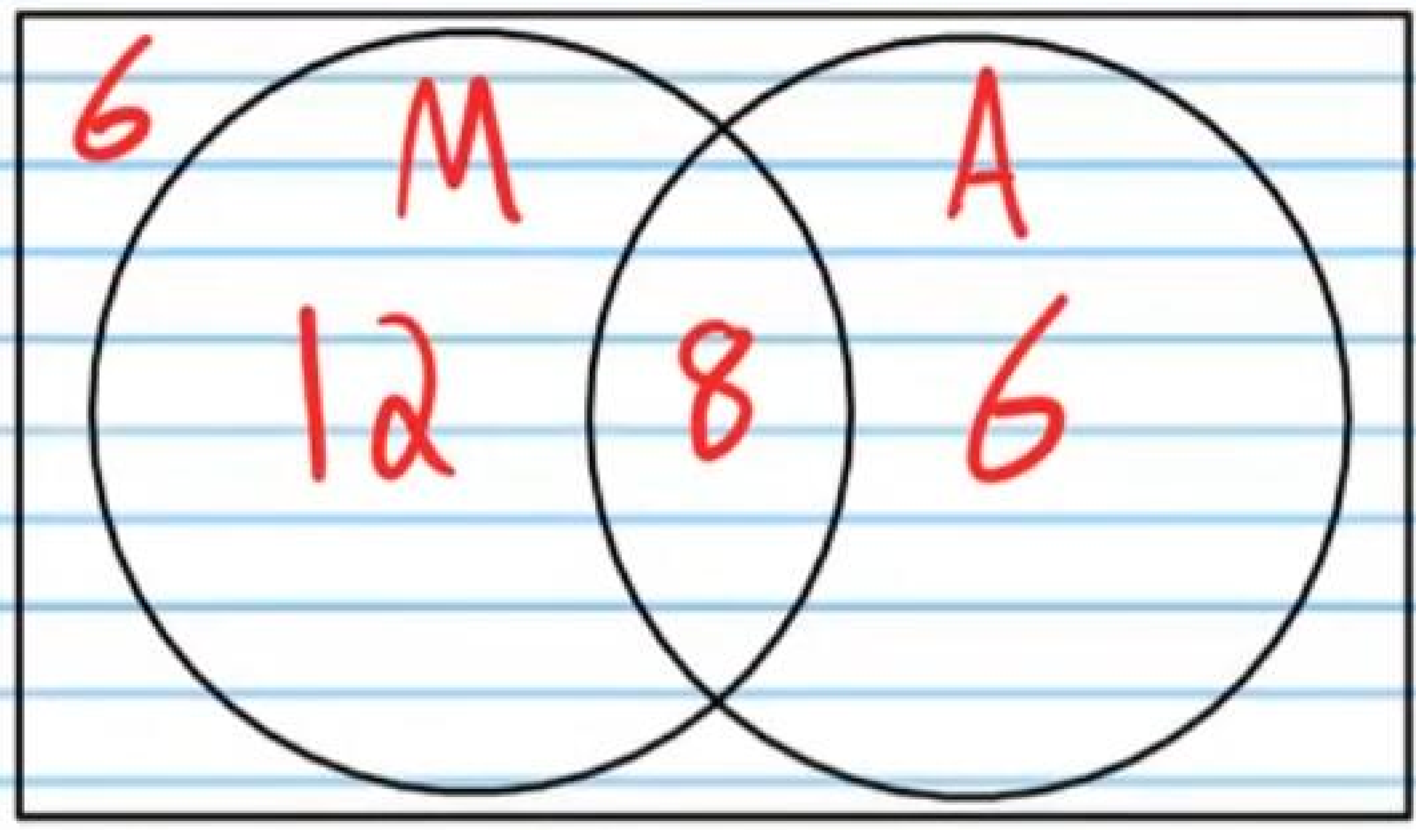
$14 - 8 = \boxed{6}$

iii) How many students like either math or art?

$12 + 8 + 6 = \boxed{26}$

iv) How many students like neither math nor art?

$32 - 26 = \boxed{6}$



v) What is the probability that a randomly selected student will like math?

$\frac{20}{32} = \boxed{\frac{5}{8}}$

vi) What is the probability that a randomly selected student will like art?

$\frac{14}{32} = \boxed{\frac{7}{16}}$

vii) What is the probability that a randomly selected student will like neither math nor art?

$\frac{6}{32} = \boxed{\frac{3}{16}}$

viii) What is the probability that a randomly selected student will like both math and art?

$\frac{8}{32} = \boxed{\frac{1}{4}}$

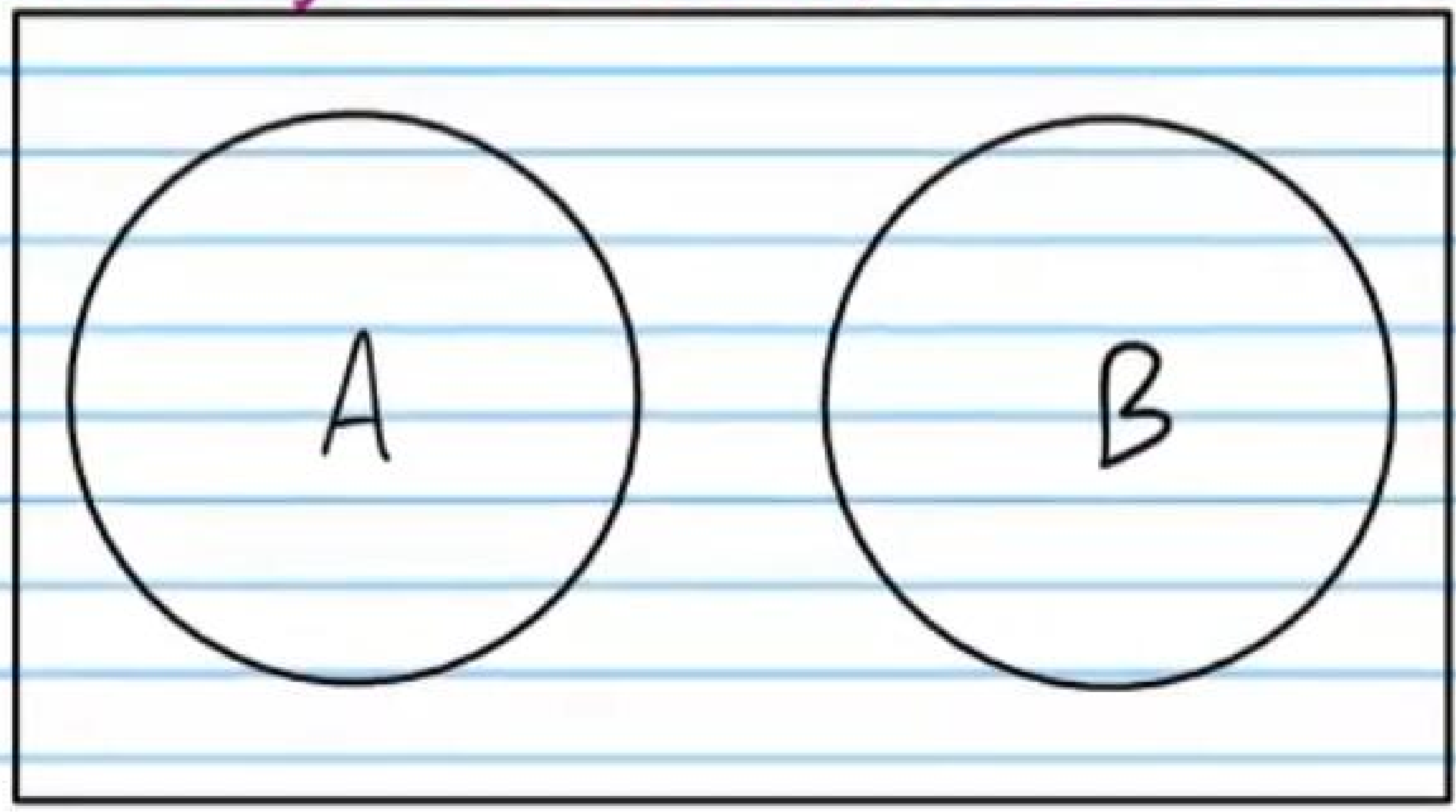
ix) What is the probability that a randomly selected student will like either math or art?

$\frac{12+8+6}{32} = \frac{26}{32} = \boxed{\frac{13}{16}}$

Scenarios

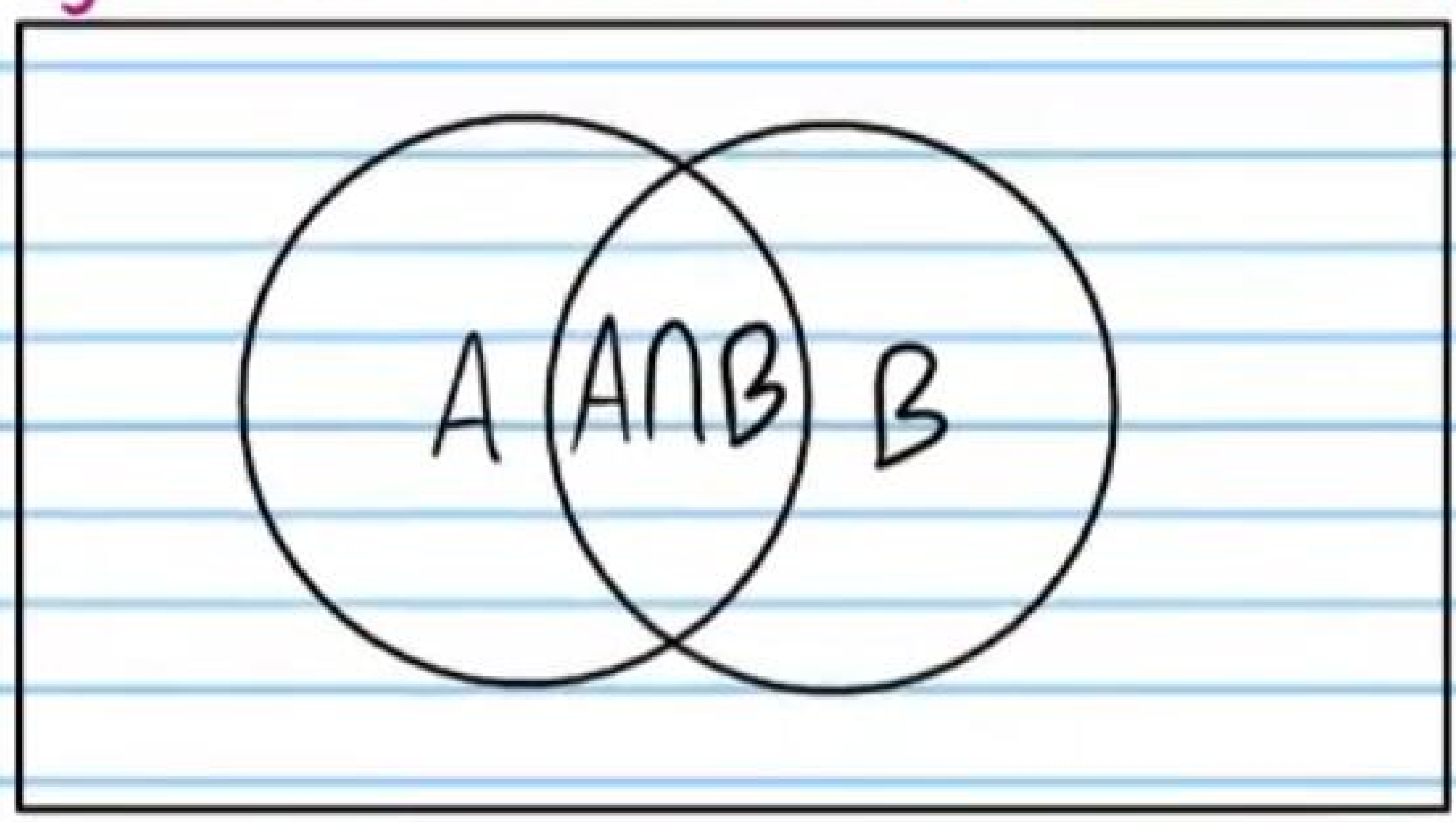
Mutually Exclusive Events: If one event occurs, then the other event cannot occur.

Venn Diagram of Mutually Exclusive Events



Example: Students can be in the band or on the football team but not both.

Venn Diagram of Events That Are Not Mutually Exclusive



Example: Alice can be in the Girl Scouts, on a soccer team, or both.

Probability Formulas

The Probability of Either "A" Or "B" if They Are Mutually Exclusive

$$P(A \cup B) = P(A) + P(B)$$

Explanation

$$P(A \cup B) = P(A) + P(B)$$

$$= \frac{\text{\# of outcomes in "A"}}{\text{Total \# of outcomes possible}} + \frac{\text{\# of outcomes in "B"}}{\text{Total \# of outcomes possible}}$$

$$= \frac{\text{\# of outcomes in "A"} + \text{\# of outcomes in "B"}}{\text{Total \# of outcomes possible}}$$

$$= P(A \cup B)$$

Calculating Probabilities of Mutually Exclusive Events

⑤ The probability that a randomly chosen animal at the zoo will be a bird is $\frac{1}{10}$. The probability that a randomly chosen animal at the zoo will be a cat is $3/20$. What is the probability that a randomly chosen animal will be either a bird or a cat?

$$P(B \cup C) = P(B) + P(C) = \frac{1}{10} + \frac{3}{20} = \frac{5}{20} = \boxed{\frac{1}{4}}$$

⑥ The probability that a randomly selected person will vote for candidate "A" is $\frac{1}{3}$. The probability that a randomly selected person will vote for candidate "B" is $1/4$. What is the probability that a randomly selected person will vote for either candidate "A" or candidate "B?"

$$P(A \cup B) = P(A) + P(B) = \frac{1}{3} + \frac{1}{4} = \boxed{\frac{7}{12}}$$

⑦ The probability that a randomly chosen phone will be a Samson phone is $2/15$. The probability that a randomly chosen phone will be an Apple phone is $7/10$. What is the probability that a randomly chosen phone will be either a Samson phone or an Apple phone?

$$P(S \cup A) = P(S) + P(A) = \frac{2}{15} + \frac{7}{10} = \frac{25}{30} = \boxed{\frac{5}{6}}$$

⑧ A student is selected at random from a particular class. The probability that the student will be 13 years old is $4/5$. The probability that the student will be 14 years old is $2/13$. What is the probability that the student will be either 13 years old or 14 years old?

$$P(13 \cup 14) = P(13) + P(14) = \frac{4}{5} + \frac{2}{13} = \frac{52}{65} + \frac{10}{65} = \boxed{\frac{62}{65}}$$

The Probability of Either "A" Or "B" if They Are Not Mutually Exclusive

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Explanation: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\frac{\text{\# of outcomes in "A"}}{\text{Total \# of outcomes possible}} + \frac{\text{\# of outcomes in "B"}}{\text{Total \# of outcomes possible}} - \frac{\text{\# of outcomes in both "A" and "B"}}{\text{Total \# of outcomes possible}}$$

\# of outcomes in "A" and "B" counted twice!

Counteracts this error

$$\frac{\text{\# of outcomes in "A"} + \text{\# of outcomes in "B"} - \text{\# of outcomes in both "A" and "B"}}{\text{Total \# of outcomes possible}}$$

$$\frac{\text{\# of outcomes in "A" only} + \text{\# of outcomes in "B" only} + \text{\# of outcomes in both "A" and "B"}}{\text{Total \# of outcomes possible}}$$

$$P(A \cup B)$$

Calculating Probabilities of Events That Are Not Mutually Exclusive

⑨ The probability that a randomly chosen animal at the zoo will be a bird is 0.18. The probability that a randomly chosen animal at the zoo will be a meat eater is 0.2. The probability that a randomly chosen animal at the zoo will be both a bird and a meat eater is 0.04. What is the probability that a randomly chosen animal will be either a bird or a meat eater?

$$P(B \cup M) = P(B) + P(M) - P(B \cap M) = 0.18 + 0.2 - 0.04 = \boxed{0.34}$$

⑩ The probability that a randomly selected person will vote for candidate "A" is 23%. The probability that a randomly selected person will vote for a new traffic law is 43%. The probability that a randomly selected person will vote for both candidate "A" and the new traffic law is 19%. What is the probability that a randomly selected person will vote for either candidate "A" or the new traffic law?

$$P(A \cup L) = P(A) + P(L) - P(A \cap L) = 0.23 + 0.43 - 0.19 = \boxed{0.47}$$

⑪ The probability that a randomly chosen phone will be a Samson phone is 0.13. The probability that a randomly chosen phone will be the color red is 0.021. The probability that a randomly chosen phone will be both a Samson phone and the color red is 0.003. What is the probability that a randomly chosen phone will be either a Samson phone or the color red?

$$P(S \cup R) = P(S) + P(R) - P(S \cap R) = 0.13 + 0.021 - 0.003 = \boxed{0.148}$$

⑫ A student is selected at random from a particular class. The probability that the student will be 13 years old is 82%. The probability that the student will have brown hair is 68%. The probability that the student will both be 13 years old and have brown hair is 59%. What is the probability that the student will either be 13 years old or have brown hair?

$$P(13 \cup B) = P(13) + P(B) - P(13 \cap B)$$
$$= 0.82 + 0.68 - 0.59$$
$$= \boxed{0.91}$$

More Challenging Probability Problems

(13) There are 7 blue marbles, 5 green marbles, and 6 red marbles in a bag. 3 marbles are chosen at random.

i) What is the probability that all three are blue?

$$P(BBB) = \frac{{}_7C_3}{{}_{18}C_3} = \boxed{\frac{35}{816}}$$

ii) What is the probability that all three are red?

$$P(RRR) = \frac{{}_6C_3}{{}_{18}C_3} = \frac{20}{816} = \boxed{\frac{5}{204}}$$

iii) What is the probability that all three are green?

$$P(GGG) = \frac{{}_5C_3}{{}_{18}C_3} = \frac{10}{816} = \boxed{\frac{5}{408}}$$

iv) What is the probability that 2 marbles will be red and 1 will be blue?

$$P(1\text{ blue }2\text{ red}) = \frac{{}_7C_1 \cdot {}_6C_2}{{}_{18}C_3} = \frac{7 \cdot 15}{816} = \frac{105}{816} = \boxed{\frac{35}{272}}$$

v) What is the probability that 2 will be green and 1 will be red?

$$P(2\text{ green }1\text{ red}) = \frac{{}_5C_2 \cdot {}_6C_1}{{}_{18}C_3} = \frac{10 \cdot 6}{816} = \frac{60}{816} = \frac{10}{136} = \boxed{\frac{5}{68}}$$

vi) What is the probability that 2 will be blue and 1 will be green?

$$P(2\text{ blue }1\text{ green}) = \frac{{}_7C_2 \cdot {}_5C_1}{{}_{18}C_3} = \frac{21 \cdot 5}{816} = \boxed{\frac{35}{272}}$$

vii) What is the probability that 2 will be green and 1 will be blue?

$$P(2\text{ green }1\text{ blue}) = \frac{{}_5C_2 \cdot {}_7C_1}{{}_{18}C_3} = \frac{10 \cdot 7}{816} = \boxed{\frac{35}{408}}$$

viii) What is the probability that 1 marble will be green, 1 will be blue, and 1 will be red?

$$P(1\text{ green }1\text{ blue }1\text{ red}) = \frac{{}_5C_1 \cdot {}_7C_1 \cdot {}_6C_1}{{}_{18}C_3} = \frac{5 \cdot 7 \cdot 6}{816}$$

$$= \boxed{\frac{35}{136}}$$

(14) There are 8 blue marbles, 4 green marbles, and 9 red marbles in a bag. 4 marbles are chosen at random.

i) What is the probability that all 4 are blue?

$$P(BBBB) = \frac{{}_8C_4}{{}_{21}C_4} = \frac{70}{5{,}985} = \frac{14}{1197} = \boxed{\frac{2}{171}}$$

ii) What is the probability that all 4 are red?

$$P(RRRR) = \frac{{}_9C_4}{{}_{21}C_4} = \frac{126}{5{,}985} = \frac{18}{855} = \frac{6}{285} = \boxed{\frac{2}{95}}$$

iii) What is the probability that all 4 are green?

$$P(GGGG) = \frac{{}_4C_4}{{}_{21}C_4} = \boxed{\frac{1}{5{,}985}}$$

iv) What is the probability that 3 marbles will be red and 1 will be blue?

$$P(3\text{ red }1\text{ blue}) = \frac{{}_9C_3 \cdot {}_8C_1}{{}_{21}C_4} = \frac{84 \cdot 8}{5{,}985} = \frac{96}{855} = \boxed{\frac{32}{285}}$$

v) What is the probability that 3 will be green and 1 will be blue?

$$P(3\text{ green }1\text{ blue}) = \frac{{}_4C_3 \cdot {}_8C_1}{{}_{21}C_4} = \frac{4 \cdot 8}{5{,}985} = \boxed{\frac{32}{5{,}985}}$$

vi) What is the probability that 2 will be green and 2 will be red?

$$P(2\text{ green }2\text{ red}) = \frac{{}_4C_2 \cdot {}_9C_2}{{}_{21}C_4} = \frac{6 \cdot 36}{5{,}985} = \boxed{\frac{24}{665}}$$

vii) What is the probability that 2 will be green and 2 will be blue?

$$P(2\text{ green }2\text{ blue}) = \frac{{}_4C_2 \cdot {}_8C_2}{{}_{21}C_4} = \frac{6 \cdot 28}{5{,}985} = \frac{56}{1995} = \boxed{\frac{8}{285}}$$

viii) What is the probability that 1 marble will be green, 1 will be blue, and 2 will be red?

$$P(1\text{ green }1\text{ blue }2\text{ red}) = \frac{{}_4C_1 \cdot {}_8C_1 \cdot {}_9C_2}{{}_{21}C_4}$$

$$= \frac{4 \cdot 8 \cdot 36}{5{,}985}$$

$$= \boxed{\frac{128}{665}}$$

The Complement Rule

$$P(A) = 1 - P(\text{not } A)$$

Explanation

$$P(A) = 1 - P(\text{not } A)$$

$$\frac{\text{\# of outcomes in "A"}}{\text{Total \# of outcomes Possible}} = 1 - \frac{\text{\# of outcomes not in "A"}}{\text{Total \# of outcomes Possible}}$$

$$= \frac{\text{Total \# of outcomes Possible}}{\text{Total \# of outcomes Possible}} - \frac{\text{\# of outcomes not in "A"}}{\text{Total \# of outcomes Possible}}$$

$$= \frac{\text{Total \# of outcomes Possible} - \text{\# of outcomes not in "A"}}{\text{Total \# of outcomes Possible}}$$

$$= \frac{\text{\# of outcomes in "A"}}{\text{Total \# of outcomes Possible}}$$

$$= P(A)$$

(15) The probability that it will not snow is 42%. What is the probability that it will snow?

$$P(S) = 1 - P(\text{not } S) = 1 - 0.42 = \boxed{0.58}$$

(16) The probability that Kyle will be accepted to his preferred college is 31%. What is the probability that he will not be accepted?

$$P(A) = 1 - P(\text{not } A)$$

$$0.31 = 1 - P(\text{not } A)$$

$$P(\text{not } A) = \boxed{0.69}$$

(17) The probability that the stock market will not crash tomorrow is 99.8%. What is the probability that it will crash?

$$P(C) = 1 - P(\text{not } C) = 1 - 0.998 = \boxed{0.002}$$

(18) The probability that a randomly selected employee will be promoted is 8%. What is the probability that a randomly selected employee will not be promoted?

$$P(p) = 1 - P(\text{not } p)$$

$$0.08 = 1 - P(\text{not } p)$$

$$P(\text{not } p) = \boxed{0.92}$$

(19) There are 6 blue marbles and 5 green marbles in a bag. 3 marbles are chosen at random.

i) What is the probability that at least one marble will be blue?

$$P(\text{at least one blue}) = 1 - P(\text{no blue})$$
$$= 1 - P(GGG)$$
$$= 1 - \frac{{}_5C_3}{{}_{11}C_3}$$
$$= 1 - \frac{10}{165}$$
$$= \frac{155}{165}$$
$$= \boxed{\frac{31}{33}}$$

ii) What is the probability that at least one marble will be green?

$$P(\text{at least 1 green}) = 1 - P(\text{no green})$$
$$= 1 - P(BBB)$$
$$= 1 - \frac{{}_6C_3}{{}_{11}C_3}$$
$$= 1 - \frac{20}{165}$$
$$= \frac{145}{165}$$
$$= \boxed{\frac{29}{33}}$$

(20) There are 8 cupcakes and 10 cookies in the refrigerator. Kevin will choose 3 of these at random.

i) What is the probability that he will choose at least one cookie?

$$P(\text{at least 1 cookie}) = 1 - P(\text{no cookies})$$

$$= 1 - P(\text{3 cupcakes})$$

$$= 1 - \frac{{}_8C_3}{{}_{18}C_3}$$

$$= 1 - \frac{56}{816}$$

$$= \frac{760}{816}$$

$$= \frac{190}{204}$$

$$= \boxed{\frac{95}{102}}$$

ii) What is the probability that he will choose at least one cupcake?

$$P(\text{at least 1 cupcake}) = 1 - P(\text{no cupcakes})$$

$$= 1 - P(\text{3 cookies})$$

$$= 1 - \frac{{}_{10}C_3}{{}_{18}C_3}$$

$$= 1 - \frac{120}{816}$$

$$= \frac{696}{816}$$

$$= \frac{232}{272}$$

$$= \frac{58}{68}$$

$$= \boxed{\frac{29}{34}}$$